

Highlights

Online Bootstrapping for Streaming Data Environment with Limited Storage by Upper-Lower-End Interval Re-sampling

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- We proposed a novel methodology called Recursively Bootstrapping at Upper-Lower Tails (RBULT), designed to address the significant limitations of conventional bootstrapping in streaming data environments.
- A two-phase recursive process that compares incoming data histograms against standard distribution shapes to automatically extend boundary intervals.
- The integration of a z-score-based outlier elimination algorithm, which ensures valid range estimates even in contaminated streaming environments where traditional methods often fail.
- Comprehensive testing across ten simulated distributions and four real-world datasets, demonstrating that RBULT significantly outperforms traditional bootstrapping in range approximation accuracy.

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ABSTRACT

Bootstrapping has been widely used as an effective non-parametric approach to estimate the population variance by resampling the entire sample with replacement. This approach does not make any assumption about the probability distribution of data, making it suitable for several real applications. Although it is very efficient and practical, the method cannot be applied to streaming data with limited memory due to the risk of data overflow. Furthermore, it is impossible to apply conventional bootstrapping to estimate the variance of data features in machine learning using the *discard-after-learn concept*, in which the learned data are entirely discarded afterwards. Recently, there have been few methods attempting to apply conventional bootstrapping to streaming data, which require pre-defined parameters and the assumption of normal distribution, which may not conform with the actual distribution of data. This paper proposes a new method called Recursively Bootstrapping at Upper-Lower Tails (RBULT) to cope with chunk-wise data in a streaming environment. Only the current incoming data is bootstrapped without any assumption on data probability distribution, information from previous chunks, and pre-defined parameters. Experimental results on ten simulated continuous streaming datasets with various probability distributions and four real-world datasets of multiple chunk sizes (50, 100, and 500) without and with outlier contamination levels (0.05% of population size) show that RBULT significantly outperform the conventional bootstrapping. In addition, the population variance estimated by RBULT is closer to the true population variance than the conventional bootstrapping method.

1. Introduction

Bootstrap method is a powerful and widely used statistical tool for estimating uncertainty from finite samples across a wide range of parameters, including standard errors, confidence intervals, and accuracy measures. Efron [2] introduced the bootstrap method as a generalisation of the Jackknife method, demonstrating that the Jackknife could be mathematically expressed as a linear approximation of the bootstrap. The empirical results showed that the proposed methods could effectively estimate the standard errors of complex estimators, establishing the foundation for modern resampling techniques. Building upon this foundation work, Efron and Tibshirani [3] further explored the basic concepts and applications of bootstrap methods for estimating standard errors, confidence intervals, and other measures of statistical accuracy. In standard error estimation, data from a population are resampled with replacement to create multiple bootstrap samples, enabling the calculation of statistics and their corresponding standard

deviations. Their results demonstrated that bootstrap estimations provided reasonably accurate and efficient alignment with theoretical density curves. Regarding confidence intervals, they empirically tested several methods. They found that bootstrap confidence intervals are more precise than standard methods, particularly when the statistical distribution deviates from normality. However, achieving accurate results typically requires a sufficiently large number of bootstrap replications. Subsequent studies have further refined the practical implementation of bootstrap methods in applied research. Carpenter and Bithell [4] presented a comprehensive guide for applying bootstrap confidence intervals in health care data, systematically addressing when, which, and how to implement these methods. They evaluated various bootstrap approaches for confidence intervals across three distinct families: pivotal, non-pivotal, and test-inversion methods. Their experimental analysis concluded that bootstrap confidence intervals serve as a suitable alternative when the assumptions underlying traditional distributional methods do not hold, such as asymptotic normality, particularly in scenarios involving small sample sizes or complex data structures.

Range estimation in univariate data sets constitutes a fundamental component of statistical analysis and computational modeling, particularly in determining the

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spread between minimum and maximum observations. [5] introduced a novel re-sampling methodology for establishing control limits in statistical process control charts, demonstrating significant advances over traditional parametric approaches across multiple chart types, including those monitoring means, ranges, standard deviations, medians, proportion defective, and defect counts. The methodology operates through a systematic process of aggregating observed data, generating multiple bootstrap samples with replacement, computing the target statistic for each sample, and deriving the control limits from the empirical percentiles of resulting distribution. This approach offers notable methodological advantages, primarily its conceptual accessibility and procedural transparency, as it relies exclusively on random sampling principles and percentile calculations, thereby eliminating complex distributional assumptions while maintaining statistical rigor and reducing technical prerequisites for implementation. The re-sampling framework represents a significant contribution to non-parametric statistical process control, offering robust alternatives to conventional approaches that may be sensitive to distributional assumptions or limited sample sizes, ultimately enhancing the ability of interpretation for practitioners across diverse analytic contexts. Palm and Nagler [6] presented an online bootstrap method for uncertainty quantification in large streams of dependent data, such as time series, directly addressing the computational and memory limitations of traditional bootstrap methods in big data and streaming environments. No restrictive assumptions about data dependence are made in their method. But some parameters must be predefined and the type of data distribution must be assumed. An auto-regressive sequence of increasingly dependent re-sampling weights was applied, allowing for $O(1)$ online updates in both memory and time for the bootstrap average and providing a significant computational advantage over the blockwise bootstrap. The authors provided rigorous theoretical validation under general conditions, including specific restrictions on time series dependence, and empirical demonstration on the effectiveness of their approach in reliable uncertainty quantification, even in the presence of complex data dependencies. Hall and Yao [7] addressed the challenge of accurately calibrating coverage for non-parametric prediction intervals, highlighting the poor performance of conventional methods, such as the standard bootstrap and Jackknife. Both the standard bootstrap and Jackknife methods often fail due to tied values in re-samples. Jackknife method, despite being unbiased for true coverage, unexpectedly increases coverage error, rendering it ineffective for this specific calibration problem. To overcome these limitations, the authors

propose two primary approaches: interpolation among adjacent order statistics and, more significantly, the use of a smoothed version of the bootstrap. These techniques were designed to reduce coverage error substantially. The smoothed bootstrap, with an empirically selected bandwidth, effectively improves the coverage accuracy and mitigates fluctuations, particularly demonstrating its strength in scenarios with small sample sizes. Basically, for conducting the bootstrap method, all data were used to repeatedly random selection to approximate a range or other statistical parameters. However, in many real-world applications, a single time for conducting the bootstrap, data are collected incrementally. This paper proposes a new method called Recursively Bootstrapping at Upper-Lower Tails (RBULT) to improve the capabilities of previous methods so as to cope with chunk-wise data in a streaming environment. Only the current incoming data is bootstrapped without any assumption on data probability distribution, information from previous chunk, and predefined parameters.

The rest of this paper is organised as follows. Section II details the studied problem and the objectives of this research. Section III presents the proposed concept of Recursively Bootstrapping at Upper-Lower Tails (RBULT) for a streaming environment. Section IV describes the experiments conducted to evaluate the performance of our method and discusses the results. Finally, Section V concludes the paper with a summary of our findings and suggestions for future work.

2. Problem Formulation and Constraints

This section discusses the studied problems under a concerned scenario and constraints which arises in various online machine learning settings.

2.1. Concerned Scenario

Sampled data continuously flow in forms of chunks into a process, which may be a learning process or a statistical analyzing process. The size of each chunk is not fixed and does not exceed the size of the working memory. Each sampled chunk contains a set of positive integers whose minimum and maximum values are unknown. The probability distribution of each integer value is not predefined and unknown. This scenario exists in various problems in machine learning. One of them is the problem of training a neural network in a streaming data environment with limited memory size by using hyper-ellipsoids [8]. Estimating the population variance of each feature of incoming data in advance can help preemptively expand the size of hyper-ellipsoids to cope with any test set, leading to improve the accuracy of classification.

2.2. Studied Problems

Suppose the size of working memory is fixed at m . Let c_i be the i^{th} incoming numeric data chunk such that $|c_i| < m$ and $i \geq 1$. After receiving chunk c_i at time $t = i$, the minimum and maximum values of population at the left and right ends are estimated. Let L denote the estimated left-end value which may be the minimum data value of chunk c_i or the minimum value computed from bootstrapping some data in c_i , depending on which one is smaller. Similar to L , let R denote the estimated right-end value which may be the maximum data value of chunk c_i or the maximum value computed from bootstrapping some data in c_i , depending on which one is larger. After estimating L and R of c_i , all data except some partial data at both ends are discarded. Those kept data are combined with the new incoming data in c_{i+1} for computing new L and R .

Figure 1 illustrates an example of the studied problem. here are three incoming chunks c_1 , c_2 , and c_3 . D denotes the working memory for accommodating c_i . Assume that the minimum incoming value is 1. At time $t = 1$, the values of L and R are estimated by bootstrapping some data of c_1 , setting $L = 3$ and $R = 19$. At time $t = 2$, the values of all data in c_2 do not exceed L and R so no bootstrapping is activated. At time $t = 3$, the values of data in c_3 exceed L and R so L and R are updated. Notice that $L = 1$ is the minimum value so no bootstrapping is applied to the left end. Only the value of R is re-estimated by bootstrapping, setting $R = 21$.

From the example in Figure 1, the objective of this study is to estimate the values of L and R so that most data in subsequent incoming chunks fall within the interval of $[L, R]$ with the minimum number of estimating the values of L and R . Therefore, the studied problems defined as follows.

1. How to estimate the population variance interval without using the entire data in set D so that the data in next incoming chunk always fall within the new interval of estimated population variance? After estimation, the almost entire data set should be discarded, making it ready to store the next incoming data chunk.
2. Theoretically, how many times on average must the data interval in set D be expanded according to the estimated population variance?

2.3. Constraints

Let $|c_i|$ represent the number of elements in a given chunk, c_i . The bin B is an interval with a left end, L , and a right end, R . The following constraints apply:

1. The probability distribution of each integer, d_j , within a chunk, c_i , is unknown.
2. The size of a chunk, $|c_i|$, may or may not be equal to the size of the next chunk, $|c_{i+1}|$. Furthermore, $|c_i| < |D|$.
3. Once all integers in a chunk, c_i , have been assigned to a bin, the chunk is completely discarded and cannot be reassigned.

3. Proposed Concept of RBULT

This study does not make any assumption about the probability distribution of incoming data chunk same as the classical bootstrap method. However, RBULT re-samples the data only in the left-end and right-end intervals. This differs from the classical bootstrap method, which re-samples data without location constraint. Due to these new re-sampling regions, two consequent problems arise. The first problem is how large the left-end and right-end intervals should be.

The size of left-end and right-end intervals are determined from the standard histogram shape of probabilistic distribution. If the heights of left-end and right-end intervals in the histogram of incoming data distribution exceed the corresponding heights of the same intervals in the standard histogram, the left-end and right-end intervals must be extended by bootstrapping the data in both intervals. Figure 2 illustrates an example of the standard histogram shown in pink color and the data histogram shown in blue color. Clearly, the height of data at the left-end and right-end intervals are exceed those of the standard histogram. There are eight intervals denoted by b_1 to b_8 .

The second consequent problem concern identifying an appropriate probabilistic distribution to solve the first problem. The appropriate probabilistic distribution is determined from the first incoming chunk, under the assumption that subsequent incoming chunks possess the same probabilistic distribution with small divergence. The experimental results in Section 6 confirm that this assumption is practically reasonable.

Suppose there are eight intervals in set B . The proposed concept has two phases. The first phase analyses the probability distribution of the first incoming data chunk and compare the standard histogram of the distribution with the actual histogram of the incoming data. If they are different, the data in b_1 and b_8 are bootstrapped to extend both boundary intervals b_1 and b_8 . All data not in b_1 and b_8 are discarded afterwards. The second phase performs the following steps:

1. Get a new data chunk and distribute the data into the corresponding intervals.

2. Compute a new center and partition new set of intervals.
3. Analyze the probability distribution of data using the new center and new set of intervals.
4. Compare the height of the standard histogram with that of the actual histogram of the incoming data. If they are different, the data in b_1 and b_8 are bootstrapped to extend both intervals.
5. Repeat steps 1-4.

Figure 3 illustrates the proposed concept of recurring online bootstrapping. A standard histogram, comprising eight bins (highlighted in pink), is generated based on a known statistical distribution, where the height of each bin represents the theoretical count of elements. The top section of Figure 3 demonstrates that when an expansion is required, the number of incoming data points in the leftmost or rightmost intervals (shaded in blue) exceeds the theoretical capacity of the standard histogram. To address this, the proposed recurring online bootstrap method incrementally extends the histogram to accommodate new data chunks. As shown in the middle section of Figure 3, after an extension is performed, the number of data points in the expanded leftmost and rightmost intervals no longer exceeds the standard height of histogram. These extensions are illustrated in the upper and lower sub-figures of the bottom section, which show right-end and left-end extensions, respectively.

4. RBULT Algorithms

This section outlines the proposed algorithms for the recurring online bootstrap in a streaming environment, with or without outlier filtering. Suppose the data in D are partitioned into eight equal intervals, $B = \{b_1, \dots, b_8\}$. Let L and R be the left-end and right-end values after bootstrapping at b_1 and b_8 respectively. Define $d^{(min)} = \min_{d \in D}(d)$ and $d^{(max)} = \max_{d \in D}(d)$. There are four main algorithms:

1. Algorithm 1: Estimating the left-end and right-end values of c_1 .
2. Algorithm 2: Estimating the left-end and right-end values of c_i for $i > 1$.
3. Algorithm 3: Bootstrapping Data in b_1 and b_8 .
4. Algorithm 4: Eliminating outlier by z-score:

The details of each algorithm are in the following subsections.

4.1. Estimating the left-end and right-end values of c_1

Algorithm 1: Estimating left-end and right-end of c_1

- Input:** (1) First data chunk, c_1 .
 (2) List P of probability density functions.
 (3) Set of empty intervals $B = \{b_1, b_2, \dots, b_8\}$ and $|b_i|$ denotes the height.
 (4) Set of empty intervals $H = \{h_1, h_2, \dots, h_8\}$ and $|h_i|$ denotes the height.

- Output:** (1) Values of left end L and right end R .
 (2) Set of intervals $B = \{b_1, b_2, \dots, b_8\}$

1. Load c_1 into D and find the probabilistic distribution best fits the data in D from list P to create standard histogram H .
3. Create 8-interval histogram B from data in D .
4. Find $d^{(min)}$ and $d^{(max)}$.
5. **While** $|b_1| > |h_1|$ **or** $|b_8| > |h_8|$ **do**
6. Bootstrap data in b_1 and b_8 to obtain L and R by Algorithm 3.
7. Set mean $\mu = \frac{L+R}{2}$ and standard deviation $\sigma = \frac{L-R}{8}$.
8. Include L and R into D .
9. Create histogram H from data in D by using μ and σ from step 7.
10. **EndWhile**
11. Discard all data in D except those in b_1 and b_8 .
12. Apply Algorithm 2 to next c_i for $i > 1$.

4.2. Estimating left-end and right-end of c_i for $i > 1$

Algorithm 2: Estimating left-end and right-end of c_i for $i > 1$

- Input:** (1) Data chunks, c_i for $i \geq 1$.
 (2) Set of intervals $B = \{b_1, b_2, \dots, b_8\}$ from Algorithm 1.
 (3) L and R from Algorithm 1.

Output: Values of left end L and right end R .

1. **While** there exists new incoming chunk c_i **do**
2. Load c_i into D .
3. Eliminate outlier data from D by Algorithm 4.
4. Set $d^{(min)} = \min_{d \in D}(d)$ and $d^{(max)} = \max_{d \in D}(d)$.
5. **If** $d^{(min)} < L$ **then** Set $L = d^{(min)}$.
6. **If** $d^{(max)} > R$ **then** Set $R = d^{(max)}$.
7. Create histogram B for data in D from mean $\mu = \frac{L+R}{2}$ and standard deviation $\sigma = \frac{R-L}{8}$.
8. Create standard histogram H of probabilistic distribution found in Algorithm 1 with μ and σ from step 7.

9. **While** $|b_1| > |h_1|$ **or** $|b_8| > |h_8|$ **do**
10. Bootstrap data in b_1 and b_8 to obtain L and R by Algorithm 3.
11. Set mean $\mu = \frac{L+R}{2}$ and standard deviation $\sigma = \frac{L-R}{8}$.
12. Include L and R into D .
13. Create histogram B from data in D using μ and σ from step 11.
14. **EndWhile**
15. Discard all data in D except those in b_1
16. **EndWhile**

4.3. Bootstrapping Data in b_1 and b_8

Algorithm 3: Bootstrapping Data in b_1 and b_8

- Input:** (1) Data in b_1 and b_8 .
(2) Heights $|b_1|$ and $|b_8|$.
(3) Predefined number of bootstrap samples N .
(4) Predefined number of bootstrap iterations M .

Output: L and R .

1. Let μ_{h_1} be the mean of data in h_1
2. and μ_{h_8} be the mean of data in h_8 .

compute value of L
3. Let $\mathcal{M} = \emptyset$ be a set of bootstrap means.
4. Let $\mathcal{S} = \emptyset$ be a set of bootstrap standard deviations.
5. **For** $1 \leq i \leq M$ **do**
6. Sample N data in b_1 with replacement and compute new mean $\hat{\mu}_{b_1}$.
7. Set $\mu_{b_1} = \frac{1}{2}(\hat{\mu}_{b_1} + \mu_{b_1})$.
8. Set $\mathcal{M} = \mathcal{M} \cup \{\mu_{b_1}\}$.
9. Sample N data in b_1 with replacement and compute new standard deviation $\hat{\sigma}_{b_1}$.
10. Set $\mathcal{S} = \mathcal{S} \cup \{\hat{\sigma}_{b_1}\}$.
11. **EndFor**
12. Compute $\mu^{(boot)} = \text{mean}(\mathcal{M})$.
13. Compute $\sigma^{(boot)} = \text{mean}(\mathcal{S})$.
14. **If** $\text{mean}(b_1) < \mu^{(boot)}$ **then**
15. Set $L = \min(b_1) - |\text{std}(b_1) - \sigma^{(boot)}|$.
16. **If** $\mu^{(boot)} < \text{mean}(b_1)$ **then**
17. Set $L = \min(b_1) - |\text{mean}(b_1) - \mu^{(boot)}|$.

compute value of R
18. Let $\mathcal{M} = \emptyset$ be a set of bootstrap means.
19. Let $\mathcal{S} = \emptyset$ be a set of bootstrap standard deviation.
20. **For** $1 \leq i \leq M$ **do**
21. Sample N data in b_8 with replacement and compute new mean $\hat{\mu}_{b_8}$.
22. Set $\mu_{b_8} = \frac{1}{2}(\hat{\mu}_{b_8} + \mu_{b_8})$.
23. Set $\mathcal{M} = \mathcal{M} \cup \{\mu_{b_8}\}$.

24. Sample N data in b_8 with replacement and compute new standard deviation $\hat{\sigma}_{b_8}$.
25. Set $\mathcal{S} = \mathcal{S} \cup \{\hat{\sigma}_{b_8}\}$.
26. **EndFor**
27. Compute $\mu^{(boot)} = \text{mean}(\mathcal{M})$.
28. Compute $\sigma^{(boot)} = \text{mean}(\mathcal{S})$.
29. **If** $\text{mean}(b_8) < \mu^{(boot)}$ **then**
30. Set $R = \max(b_8) + |\text{std}(b_8) - \sigma^{(boot)}|$.
31. **If** $\mu^{(boot)} < \text{mean}(b_8)$ **then**
32. Set $R = \max(b_8) + |\text{mean}(b_8) - \mu^{(boot)}|$.

Define $\text{mean}(\mathcal{M})$ as a function to compute the average of data in a set $\mathcal{M}$. Also define $\text{std}(\mathcal{M})$ as a function to compute the standard deviation of data in a set $\mathcal{M}$, as well.

4.4. Eliminating Outlier Data

Algorithm 4: Eliminating outlier by z-score:

- Input:** (1) Data set D .
(2) A constant τ .

Output: Data set D without outliers.

1. Compute mean μ of D .
2. Compute standard deviation σ of D .
3. **For** each $d \in D$ **do**
4. Compute $z = \frac{|d-\mu|}{\sigma}$.
5. **If** $z > \tau$ **then**
6. $D = D - \{d\}$.
7. **EndFor**

5. Theoretical Analysis

The minimum and maximum numbers of expansion for the values of L and R can be theoretically analysed in terms of probability as addressed in the following theorems. Suppose the size of population $\mathcal{P}$ is known to be N and the size of each incoming chunk is l . Each datum has no duplicate. Let $p^{(min)} = \min(\mathcal{P})$ and $p^{(max)} = \max(\mathcal{P})$ be the minimum and maximum data of the population $\mathcal{P}$, respectively.

Theorem 1. Assume that a chunk of l data is picked one at a time without replacement from the population $\mathcal{P}$ with size N . The probability of having $p^{(min)}$ and $p^{(max)}$ in the first incoming chunk is $\frac{l(l-1)}{N(N-1)}$.

Proof. When $p^{(min)}$ and $p^{(max)}$ are in the first chunk, the rest of $l-2$ data must be randomly picked from $N-2$ data of population. The number of ways to pick $l-2$ data from $N-2$ data is

$$w = \binom{N-2}{l-2}$$

Each picked $l - 2$ data with $p^{(min)}$ and $p^{(max)}$ can be combined with other $N - l$ data. There are $(N - l)!$ possible permutations of $N - l$ data. Therefore, the total possible number of chunks C having $p^{(min)}$ and $p^{(max)}$ is

$$\begin{aligned} c &= w \times (N - l)! \\ &= \binom{N - 2}{l - 2} (N - l)! \\ &= \frac{(N - 2)!(N - l)!}{(l - 2)!(N - 2 - l + 2)!} \\ &= \frac{(N - 2)!}{(l - 2)!} \end{aligned}$$

The total number of possible positions of $p^{(min)}$ and $p^{(max)}$ in the first chunk is $l!$. Therefore, all possible ways to have $p^{(min)}$ and $p^{(max)}$ in the first chunk is $\frac{l!(N - 2)!}{(l - 2)!} = l(l - 1)(N - 2)!$. All possible sequences of data in the population is $N!$. Thus, the probability of having $p^{(min)}$ and $p^{(max)}$ in the first incoming chunk is $\frac{l(l - 1)(N - 2)!}{N!} = \frac{l(l - 1)}{N(N - 1)}$. $\square$

Theorem 2. Assume that a chunk of l data is picked one at a time without replacement from population $\mathcal{P}$ with size N . The probability of having $p^{(min)}$ and $p^{(max)}$ in the i^{th} incoming chunk for $i \geq 1$ is $\frac{l(l - 1)}{N(N - 1)}$.

Proof. Let S denote the sample space of events of generating L data chunks from the observations in $\mathcal{P}$ by which $N = l \cdot L$. The number of events in S denoted by $|S|$, is the number of permutation of N data divided by the permutation of l elements of L data chunks. $|S|$ is mathematically expressed by

$$|S| = \frac{N!}{(l!)^L}. \quad (1)$$

Let $E_i \subset S$ denote the set of events where both $p^{(min)}$ and $p^{(max)}$ are contained within the i^{th} data chunk. Applying the product rule of combinatorics, the total number of such events, denoted by $|E_i|$, is mathematically expressed as:

$$|E_i| = \binom{N - 2}{l - 2} \frac{(N - l)!}{(l!)^{L - 1}}, \quad (2)$$

where $\binom{N - 2}{l - 2}$ represents the number of ways to select the remaining $l - 2$ observations for the i^{th} chunk from the $N - 2$ available data (excluding $p^{(min)}$ and $p^{(max)}$), $\frac{(N - l)!}{(l!)^{L - 1}}$ denotes the number of ways to partition the remaining

$N - l$ observations into the other $L - 1$ data chunks, each of size l .

Based on Equations 1 and 2, the probability of $p^{(min)}$ and $p^{(max)}$ being located within the i^{th} incoming chunk, denoted by $P(E_i)$, is derived as follows:

$$\begin{aligned} P(E_i) &= \frac{|E_i|}{|S|} \\ &= \frac{\binom{N - 2}{l - 2} \frac{(N - l)!}{(l!)^{L - 1}}}{\frac{N!}{(l!)^L}} \\ &= \frac{\left(\frac{(N - 2)!}{(l - 2)!(N - l)!} \right) \left(\frac{(N - l)!}{(l!)^{L - 1}} \right)}{\frac{N!}{(l!)^L}} \\ &= \left(\frac{(N - 2)!}{(l - 2)!(N - l)!} \right) \left(\frac{(N - l)!}{(l!)^{L - 1}} \right) \left(\frac{l!(l!)^{L - 1}}{N!} \right) \\ &= \frac{(N - 2)!(N - l)!l!(l!)^{L - 1}}{(l - 2)!(N - l)!(l!)^{L - 1}N!}, \end{aligned}$$

finally, we obtain that

$$P(E_i) = \frac{l(l - 1)}{N(N - 1)}. \quad (3)$$

Therefore, the probability of having $p^{(min)}$ and $p^{(max)}$ in the i^{th} incoming chunk for $i \geq 1$ is $\frac{l(l - 1)}{N(N - 1)}$. $\square$

Corollary 1. Assuming that the joint occurrence of the minimum and maximum observations is restricted to a single chunk, the expected index of such a chunk is given by $E[X] = \frac{1}{2} \left(\frac{l(l - 1)(L)(L + 1)}{N(N - 1)} \right)$ for the total number of L data chunks.

Proof. Let $X \in \{1, 2, \dots, L\}$ be a discrete random variable representing the index of the chunk containing both $p^{(min)}$ and $p^{(max)}$. Given the assumption that this event must occur in exactly one of the L chunks, and note that the probability is uniform across all chunks ($P(E_i) = \frac{l(l - 1)}{N(N - 1)}$ under this restriction), the expectation is calculated as follows:

$$\begin{aligned} E[X] &= \sum_{i=1}^L i \cdot P(X = i) \\ &= \sum_{i=1}^L i \cdot \frac{l(l - 1)}{N(N - 1)} \\ &= \left(\frac{L(L + 1)}{2} \right) \left(\frac{l(l - 1)}{N(N - 1)} \right) \end{aligned}$$

$$= \frac{1}{2} \left(\frac{l(l-1)(L)(L+1)}{N(N-1)} \right).$$

Finally, we obtain:

$$E[X] = \frac{1}{2} \left(\frac{l(l-1)(L)(L+1)}{N(N-1)} \right) \quad (4)$$

□

6. Experiments

6.1. Experimental design

This study comprehensively evaluates the range approximation method across two types of one-dimensional data populations: simulated data and real-world data. As detailed in Table 1, ten datasets were generated from five established statistical distributions, including *F*, *Chi-square*, *Normal*, *Wald*, and *Weibull*. Each has distinct parameter settings. For the *F* distribution, two configurations, $F(5, 10)$ and $F(5, 20)$, were considered to highlight differences in their statistical characteristics. Although both distributions share the same numerator parameter, $v_1 = 5$, increasing the denominator parameter, v_2 , from 10 to 20 results in notable changes in their properties. For the *Wald* distribution, a smaller λ value exhibits a more spread-out distribution with a range of [0.0323, 15.1948]. In contrast, a larger λ value results in a more concentrated distribution around the mean, with a narrower range of [0.1090, 12.8305]. For the *Normal* distributions, we used the same mean of zero while varying the variances to $\sigma^2 = 1$ and $\sigma^2 = 16$, with ranges from -3.6943 to 4.2593 and -14.2604 to 18.0044, respectively. Both *Normal* distributions share the same mean but have significantly different variances, allowing us to investigate the impact of different spread patterns under the same mean parameter, leading to distinct characteristics. For the *Chi-square* distribution, $\chi^2(\nu = 2)$ and $\chi^2(\nu = 10)$ were simulated. The $\chi^2(\nu = 2)$ distribution shows a highly right-skewed shape with its peak near zero and a range of [0.0000, 21.9290], while $\chi^2(\nu = 10)$ displays a more symmetric shape, with a broader range of [0.7468, 33.6357], reflecting greater spread and variability. For the *Weibull* distribution, *shape* parameters of 1 and 5 were used to represent distinct behavioural characteristics. *Weibull* with *shape* = 1 behaves the same as an exponential distribution with a monotonically decreasing pattern and a wide range of [0.1090, 12.2323], characterised by a constant hazard rate, making it suitable for modelling random failures and exponential decay processes. In contrast, *Weibull* with *shape* = 5 presents a bell-shaped, nearly symmetric distribution with a narrower range

of [0.1209, 1.5980], featuring increasing failure rates over time and a more concentrated probability density. These significant differences in ranges and probability density distributions reflect their reliability engineering and failure analysis applications. In addition, we set the number of simulated data to 10,000 for all datasets, treating this as the population size for constructing the streaming data environment. The distributions of all experimental data for the entire population and for the random samples are illustrated in Figures 4 - 5, using violin plots. The blue distribution represents the complete simulated population ($N = 10,000$), while the red distribution represents a 30% of random sample drawn from the population.

Additionally, we conducted more experiments on a sample dataset containing outliers. Outlier generation was achieved in two steps. First, the mean and standard deviation of the pre-defined population data were computed. Subsequently, a contamination rate (0.05% in this study) was used to determine the exact number of outliers to be added to the sample dataset. Outliers were created by generating new values from a normal distribution at a significant distance from the original data mean, defined as a multiple of the standard deviation, set to 10 in this study. Specifically, half of the outliers were created on the left side (mean minus a multiple of the standard deviation). In contrast, the remaining were created on the right side (mean plus a multiple of the standard deviation). These newly generated outlier values were then combined with the sample data chunk to produce a new data chunk containing a specified proportion of outliers.

In our experiments on real-world scenarios, we examined four datasets representing various economic sectors, as summarised in Table 2. These datasets were selected to evaluate the proposed methodology across different real-world applications and value distributions. The first dataset is the laptop prices in *Euros* [11], with values ranging from 174.00 to 6,099.00. This dataset comprehensively represents price distribution in the technology retail sector. The second dataset is electronic sales transactions [12], with values ranging from 20.75 to 11,396.80. This extensive range captures the full spectrum of electronic retail activity, from minor accessory purchases to significant investments in premium electronic equipment. The third dataset examines e-commerce sales data [13], with transaction values ranging from 100.30 to 9,995.62. This range reflects the diverse nature of online retail transactions in the digital marketplace. The distribution of the dataset provides valuable insights into e-commerce transaction patterns and consumer spending behaviours. The fourth dataset is the tourism and economic impact dataset

Table 1

Description of statistical distribution's parameters, minimum and maximum values of the simulation scenario.

No.	Distribution	Minimum	Maximum
1	$F(v_1 = 5, v_2 = 10)$	0.0083	52.4746
2	$F(v_1 = 5, v_2 = 20)$	0.0219	127.5459
3	$Wald(\mu = 1.0, \lambda = 0.5)$	0.0323	15.1948
4	$Wald(\mu = 1.0, \lambda = 2.0)$	0.1090	12.8305
5	$N(\mu = 0.0, \sigma^2 = 1.0)$	-3.6943	4.2593
6	$N(\mu = 0.0, \sigma^2 = 16.0)$	-14.2604	18.0044
7	$\chi^2(v = 2)$	0.0000	21.9290
8	$\chi^2(v = 10.0)$	0.7468	33.6357
9	$Weibull(shape = 1.0)$	0.1090	12.2323
10	$Weibull(shape = 5.0)$	0.1209	1.5980

Note: The values in the Minimum and Maximum columns represent the population minimum and maximum values of the corresponding distribution.

Table 2

Description of minimum and maximum values of the real-world scenario.

No.	Dataset	Minimum	Maximum
1	Laptop prices	174.00	6,099.00
2	Electronic sales	20.75	11,396.80
3	E-commerce sales	100.30	9,995.62
4	World tourism economy	0.1578	28.1923

Note: The values in the Minimum and Maximum columns represent the population minimum and maximum values of the corresponding dataset.

[14], which uses tourism expenditures in *U.S.* dollars. The range of this variable is from 0.1578 to 28.1923. This dataset's scale and distribution characteristics offer an important perspective on the applicability of the proposed methodology to macroeconomic indicators. The histograms of the real-world datasets are shown in Figure 6.

6.2. Experimental results

In this study, we established a streaming data chunk environment to evaluate our proposed methods. Given a population $\mathcal{P}$ with a range of $[L_{\mathcal{P}}, R_{\mathcal{P}}]$, we randomly selected a set of samples $\mathbf{X} = \{x_i \in \mathcal{P} \mid \text{for } i = 1, 2, \dots, n \text{ and } n \in \mathbb{N}\}$ from the population. These samples were used to create streaming data chunks $\mathbf{c}_i$. We empirically examined three bootstrapping-based methods for range approximation. The first method, method *a*, implements the proposed algorithms without noise filtering for the

bootstrapping-based range approximation. The second method, method *b* referred to as double online bootstrap, extends method *a* by approximating the minimum and maximum of bootstrapping datasets by performing the bootstrapping procedures detailed in Algorithms 2 and 3. The third method, method *c*, represents the traditional bootstrapping approach where data from streaming chunks are accumulated before performing bootstrapping. The method *d* is the online bootstrap with noise filtering. To evaluate the performance of these methods, we employed the range error, which quantifies the accuracy of the range approximation. The range error is calculated as follows:

$$range_error = pop_range - est_range,$$

where $pop_range = R_{\mathcal{P}} - L_{\mathcal{P}}$, denotes the population range, and $est_range = R - L$ denotes the estimated range.

6.2.1. Results on the streaming environment from simulated data scenario

The experimental results of four methods for range approximation with the equally pre-defined chunk sizes of $|\mathbf{c}_i| = 50$ and $|\mathbf{c}_i| = 500$ for simulated data without noise are given in Tables 3 and 4. The methods are: Online Bootstrap (*a*), Double Online Bootstrap (*b*), Traditional Bootstrap (*c*), and Online Bootstrap with outlier filtering (*d*). A key metric for evaluating these methods is the Range error, which measures the difference between the estimated range and the true population range. A smaller range error indicates better performance. The results in Tables 3 show that the proposed online bootstrap methods (*a*, *b*, *d*) are more effective at approximating the population range than the traditional bootstrap method (*c*) in a streaming data environment. The best performance is typically observed with the online bootstrap approach that incorporates outlier elimination (*d*), as it more effectively handles streaming data and yields a more accurate range estimate.

For comparison on online bootstrap methods, the Online Bootstrap (*a*) and the Online Bootstrap with outlier elimination (*d*) exhibit comparable performance; however, the method with outlier elimination often has a slight edge in accuracy, as reflected by its marginally lower range error in most cases. The Double Online Bootstrap (*b*) also performs competitively but does not consistently outperform the single online bootstrap methods. To further assess statistical significance, paired Student t-tests or Wilcoxon signed-rank tests were used to compare the performance of the Online Bootstrap with outlier elimination (*d*) with that of the

other three methods across all datasets. The null hypothesis was given as follows: H_0 : "There is no significant difference in the range error mean between Method d and the compared method" while the alternative hypothesis was, H_1 : "There is a significant difference in range error mean between method d and the compared method". The results show that, for the majority of the datasets, the H_0 is rejected, indicating that method d provides a statistically significant improvement in range approximation compared to the other methods, particularly the traditional bootstrap method. For Table 4, across all datasets, the online bootstrap methods (a , b , and d) consistently outperform the Traditional Bootstrap (c). For instance, in the $F(5, 10)$ distribution, the traditional bootstrap's range error is 26.40, while the online bootstrap methods maintain a substantially lower error of 24.17. Notably, the Online Bootstrap with outlier elimination (d) most frequently achieves the lowest range error, particularly for the F , $Chi - square$, and $Weibull$ distributions, where it provides the most precise range estimates.

6.2.2. Results on the streaming environment from simulated data with an outlier contaminating scenario

Table 5 displays the performance of four methods for range approximation with the chunk size of $|c_i| = 50$, when the simulated data is mixed with noise. The results highlight a significant difference in how the methods handle data contamination. Methods a and b (Online Bootstrap and Double Online Bootstrap) frequently fail to produce valid range approximations for non-negative distributions (e.g., the F and χ^2) distributions. This failure is indicated by " n/a " in the Range error, as the estimated range becomes negative and the error cannot be computed. The traditional bootstrap (method c) also fails to produce valid results for these distributions. In contrast, the Online Bootstrap with Outlier Filtering (method d) consistently provides valid and accurate range estimates, even for the most challenging non-negative distributions. For example, in the $F(5, 10)$ and $F(5, 20)$ distributions, method d provides low range errors of 0.02 and 77.20, respectively, while the other methods are unable to compute valid errors. For distributions where all methods yield valid estimates (e.g., $Wald$ and $Weibull$ distributions), the online methods (a , b , d) generally have smaller range errors than the traditional method (c). The Online Bootstrap with Outlier Filtering (d) often achieves the lowest error, with range errors of 0.10 for the $Wald(1.0, 0.5)$ distribution and 0.00 for the $Weibull(1)$ distribution.

For Table 6, the methods a (Online Bootstrap), b (Double Online Bootstrap), and c (Traditional Bootstrap) frequently fail to provide a valid range approximation for distributions with a non-negative population range, such as the F and χ^2 distributions. In these cases, the estimated range includes negative values, making the error calculation invalid, which highlights the lack of robustness in a noisy, streaming environment. In contrast, the Online Bootstrap with Outlier Filtering (Method d). It consistently provides valid range estimates for all distributions. For challenging non-negative distributions, it yields very low range errors, such as 77.20 for the $F(5, 20)$ distribution, while other methods fail. For distributions where all methods produce valid results (e.g., $Wald$ and $Weibull$), the online methods generally outperform the traditional method. The outlier-filtered approach (Method d) often achieves the lowest range error.

6.2.3. Results on the streaming environment from a real-world data scenario

Tables 7 and 8 show the performance of four range approximation methods on real-world datasets for the chunk sizes $|c_i| = 50$ and $|c_i| = 100$, respectively. For the Laptop Prices and World Tourism Economy datasets, the online bootstrap methods (a , b , and d) consistently and significantly outperform the traditional bootstrap (c). The online methods achieve identical and substantial lower range errors, such as 17.90 for Laptop Prices and 2.64 for World Tourism Economy ($|c_i| = 50$). The traditional bootstrap's errors are much higher, namely 945.14 and 1,021.8 for both ($|c_i| = 50, 100$), respectively. For Electronic sales and E-commerce datasets, the online methods perform comparably to, or even better than, traditional bootstrap (c), particularly for the Electronic Sales dataset with a chunk size of $|c_i| = 100$. Furthermore, the online bootstrap with outlier filtering (d) produces the minimum value of range error of 3 for the E-commerce dataset with $|c_i| = 100$, indicating that it outperforms the other methods.

7. Conclusions

In this work, we introduce the concept of a recurring online bootstrap for handling range approximation in streaming data environments. We propose online bootstrap algorithms with and without outlier filtering, and also a double bootstrap approach for estimating the minimum and maximum values of incoming chunks. The proposed online methods offer a significant advantage over the traditional approach for range approximation under streaming constraints. Operating in scenarios with unknown data distributions and a discarded data stream, the recurring online bootstrap effectively adapts

Table 3Estimated results for a chunk size $|c| = 50$ in simulated datasets without noise.

No.	Distribution	Population range (<i>pop_range</i>)	Estimated range (<i>est_range</i>)	Range error (<i>rage_error</i>)	Reject H_0
1.	$F(5, 10)$	[0.0083, 52.47]	[0.0066, 28.30] _a [0.0067, 28.30] _b [0.0092, 26.07] _c [0.0067, 28.30] _d	24.17 24.17 26.40 24.17	False _t True _t True _t -
2.	$F(5, 20)$	[0.0219, 127.55]	[0.0374, 39.56] _a [0.0374, 39.56] _b [0.0404, 37.23] _c [0.0374, 39.56] _d	88.00 88.00 90.35 88.00	False _t True _t True _t -
3.	$Wald(1.0, 0.5)$	[0.0323, 15.19]	[0.0355, 15.10] _a [0.0414, 15.10] _b [0.0415, 14.59] _c [0.0344, 15.10] _d	0.09 0.10 0.61 0.09	False _t True _t False _t -
4.	$Wald(1.0, 2.0)$	[0.1090, 12.83]	[0.1279, 6.47] _a [0.1280, 6.47] _b [0.1298, 5.86] _c [0.1254, 6.47] _d	6.38 6.38 6.99 6.38	False _w False _w False _w -
5.	$N(0, 1)$	[-3.6943, 4.26]	[-3.6943, 3.93] _a [-3.6943, 3.93] _b [-3.4916, 3.69] _c [-3.6943, 3.93] _d	0.33 0.33 0.77 0.33	False _t False _t True _t -
6.	$N(0, 4)$	[-14.26, 18.00]	[-13.93, 18.00] _a [-13.93, 18.00] _b [-13.71, 16.74] _c [-13.93, 18.00] _d	0.33 0.33 1.81 0.33	False _t False _t True _t -
7.	$\chi^2(10)$	[0.7468, 33.64]	[1.0809, 30.55] _a [1.0809, 30.55] _b [1.2533, 29.95] _c [1.0809, 30.55] _d	3.42 3.42 4.19 3.42	False _t False _t False _t -
8.	$\chi^2(2)$	[0.0000, 21.93]	[-0.0008, 17.10] _a [0.0001, 17.10] _b [0.0002, 16.18] _c [-0.0057, 17.10] _d	4.82 4.82 5.75 4.82	True _t True _t True _t -
9.	$Weibull(1)$	[0.0000, 12.23]	[-0.0010, 12.23] _a [0.0001, 12.23] _b [0.0002, 10.47] _c [-0.0087, 12.23] _d	-0.00 0.00 1.77 -0.01	False _t True _t False _t -
10.	$Weibull(5)$	[0.1209, 1.60]	[0.1812, 1.56] _a [0.1812, 1.56] _b [0.1988, 1.55] _c [0.1812, 1.56] _d	0.10 0.10 0.12 0.10	False _t False _t False _t -

- Estimated range with the subscripts a, b, and d stand for online bootstrap, double online bootstrap, traditional bootstrap, and online bootstrap with noise filtering estimators, respectively.

- Range error, indicated by the bold typeface, shows the minimum error. Its positive or negative values suggest whether the estimated range is narrower or wider than the true population range, respectively.

- Hypothesis test: H_0 : "There is no significant difference in the mean range error between Method d and the compared method over the specified time period.

to incoming data chunks by incrementally expanding its bin set. The empirical results demonstrate that the traditional bootstrap often struggles with outliers or non-negative data, in some cases failing to provide valid range approximations. In contrast, the online methods, particularly the one incorporating outlier

filtering, consistently deliver superior accuracy and robustness. These findings confirm the priori assumption that adaptive, real-time bootstrap strategies are fundamentally better suited for handling the challenges of streaming data than a static, traditional bootstrap method.

Table 4Estimated results for a chunk size $|c| = 500$ in simulated datasets.

No.	Distribution	Population range (<i>pop_range</i>)	Estimated range (<i>est_range</i>)	Range error (<i>rage_error</i>)	Reject H_0
1.	$F(5, 10)$	[0.0083, 52.47]	[0.0066, 28.30] _a [0.0067, 28.30] _b [0.0092, 26.07] _c [0.0067, 28.30] _d	24.17 24.17 26.40 24.17	False _t True _t True _t -
2.	$F(5, 20)$	[0.0219, 127.55]	[0.0374, 39.56] _a [0.0368, 15.10] _b [0.0404, 37.54] _c [0.0374, 39.56] _d	88.00 88.00 90.03 88.00	False _t True _t True _t -
3.	$Wald(1.0, 0.5)$	[0.0323, 15.19]	[0.0368, 15.10] _a [0.0368, 15.10] _b [0.0384, 14.62] _c [0.0364, 15.10] _d	0.10 0.10 0.58 0.10	False _w False _w False _w -
4.	$Wald(1.0, 2.0)$	[0.1090, 12.83]	[0.1270, 6.47] _a [0.1810, 6.47] _b [0.1298, 5.88] _c [0.1273, 6.47] _d	6.38 6.43 6.99 6.38	False _w True _w False _w -
5.	$N(0, 1)$	[-3.6943, 4.26]	[-3.6943, 3.93] _a [-3.6943, 3.93] _b [-3.5066, 3.69] _c [-3.6943, 3.93] _d	0.33 0.33 0.76 0.33	True _t True _t True _t -
6.	$N(0, 4)$	[-14.26, 18.00]	[-13.93, 18.00] _a [-13.93, 18.00] _b [-13.71, 16.94] _c [-13.93, 18.00] _d	0.33 0.33 1.64 0.33	False _t False _t True _t -
7.	$\chi^2(2)$	[0.7468, 33.64]	[1.0809, 30.55] _a [1.0809, 30.55] _b [1.2471, 29.92] _c [1.0809, 30.55] _d	3.42 3.42 4.21 3.42	False _w False _w False _w -
8.	$\chi^2(10)$	[0.0000, 21.93]	[0.0001, 17.10] _a [0.0001, 17.10] _b [0.0002, 16.18] _c [-0.0003, 17.10] _d	4.83 4.83 5.75 4.82	False _w True _t False _w -
9.	$Weibull(1)$	[0.0000, 12.23]	[-0.0025, 12.23] _a [0.0021, 12.23] _b [0.0002, 10.47] _c [0.0001, 12.23] _d	-0.00 0.00 1.76 0.00	False _w True _t False _w -
10.	$Weibull(5)$	[0.1209, 1.60]	[0.1812, 1.56] _a [0.2351, 1.47] _b [0.2002, 1.55] _c [0.1812, 1.56] _d	0.10 0.25 0.12 0.10	False _t True _w True _w -

- Estimated range with the subscripts SaS, SbS, and SdS stand for online bootstrap, double online bootstrap, traditional bootstrap, and online bootstrap with noise filtering estimators, respectively.

- Range error, indicated by the bold typeface, shows the minimum error. Its positive or negative values indicate whether the estimated range is narrower or wider than the true population range, respectively.

- Hypothesis test: H_0 : "There is no significant difference in the range error mean between method d and the compared method over the specified time period.

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Table 5Estimated results for chunk size $|c| = 50$ with mixing outlier data in simulated datasets.

No.	Distribution	Population range (<i>pop_range</i>)	Estimated range (<i>est_range</i>)	Range error (<i>rage_error</i>)	Reject H_0
1.	$F(5, 10)$	[0.0083, 52.47]	$[-22.66, 52.47]_a$ $[0.0800, 52.47]_b$ $[-19.44, 44.62]_c$ $[0.0290, 52.47]_d$	n/a 0.07 n/a 0.02	n/a True _t n/a -
2.	$F(5, 20)$	[0.0219, 127.55]	$[-20.66, 50.35]_a$ $[0.0232, 50.35]_b$ $[-22.99, 43.23]_c$ $[0.0219, 50.35]_d$	n/a 77.20 n/a 77.20	n/a True _t n/a -
3.	$Wald(1.0, 0.5)$	[0.0323, 15.19]	$[-13.17, 15.10]_a$ $[0.0400, 15.10]_b$ $[-11.73, 14.89]_c$ $[0.0363, 15.10]_d$	n/a 0.10 n/a 0.10	n/a True _t n/a -
4.	$Wald(1.0, 2.0)$	[0.1090, 12.83]	$[-6.25, 8.76]_a$ $[0.1280, 8.76]_b$ $[-5.28, 8.50]_c$ $[-5.92, 8.76]_d$	n/a -13.11 n/a n/a	n/a - n/a n/a
5.	$N(0, 1)$	[-3.6943, 4.26]	$[-9.97, 10.16]_a$ $[-9.97, 10.16]_b$ $[-8.83, 9.16]_c$ $[-9.97, 10.16]_d$	-12.17 -12.17 -10.67 -12.17	False _t False _t True _t -
6.	$N(0, 4)$	[-14.26, 18.00]	$[-42.90, 40.43]_a$ $[-42.90, 40.43]_b$ $[-38.72, 38.46]_c$ $[-42.90, 40.43]_d$	-51.05 -51.05 -44.92 -51.05	False _t False _t True _t -
7.	$\chi^2(2)$	[0.0000, 21.93]	$[-17.85, 24.33]_a$ $[-17.85, 24.33]_b$ $[1.2471, 29.92]_c$ $[1.0809, 30.55]_d$	n/a n/a -16.85 -2.41	n/a n/a False _t -
8.	$\chi^2(10)$	[0.7468, 33.64]	$[-33.59, 57.96]_a$ $[-33.59, 57.96]_b$ $[2.73, 18.76]_c$ $[2.63, 20.73]_d$	n/a n/a 16.86 14.78	n/a n/a True _w -
9.	$Weibull(1)$	[0.0000, 12.23]	$[0.0141, 5.78]_a$ $[0.0001, 12.23]_b$ $[-8.34, 11.35]_c$ $[0.0001, 12.23]_d$	6.47 0.00 n/a 0.00	False _t True _t n/a -
10.	$Weibull(5)$	[0.1209, 1.60]	$[-1.20, 3.22]_a$ $[-1.20, 3.22]_b$ $[-0.96, 3.08]_c$ $[-1.20, 3.22]_d$	-2.95 -2.95 0.12 -2.95	n/a n/a True _t -

- Estimated range with the subscripts SaS, SbS, and SdS stand for online bootstrap, double online bootstrap, traditional bootstrap, and online bootstrap with noise filtering estimators, respectively.

- Range error, indicated by the bold typeface, shows the minimum error. Its positive or negative values indicate whether the estimated range is narrower or wider than the true population range, respectively.

- Hypothesis test: H_0 : "There is no significant difference in the range error mean between method d and the compared method over the specified time period.

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Table 6Estimated results for the chunk size $|c| = 500$ with mixing outlier data in simulated datasets.

No.	Distribution	Population range (<i>pop_range</i>)	Estimated range (<i>est_range</i>)	Range error (<i>rage_error</i>)	Reject H_0
1.	$F(5, 10)$	[0.0083, 52.47]	$[-22.66, 52.47]_a$ $[0.0412, 52.47]_b$ $[-19.88, 44.05]_c$ $[0.0300, 52.47]_d$	n/a 0.03 n/a 0.02	n/a True_t n/a -
2.	$F(5, 20)$	[0.0219, 127.55]	$[-26.66, 50.35]_a$ $[-26.66, 50.35]_b$ $[-23.01, 43.32]_c$ $[0.0219, 50.35]_d$	n/a n/a n/a 77.20	n/a n/a n/a -
3.	$Wald(1.0, 0.5)$	[0.0323, 15.19]	$[-13.16, 15.10]_a$ $[-13.16, 15.10]_b$ $[-11.09, 14.92]_c$ $[0.0400, 15.10]_d$	n/a n/a n/a -1.96	n/a n/a n/a -
4.	$Wald(1.0, 2.0)$	[0.1090, 12.83]	$[-6.25, 8.76]_a$ $[0.1810, 8.76]_b$ $[-5.27, 8.52]_c$ $[0.1270, 8.76]_d$	n/a 4.14 n/a 4.09	n/a True_w n/a -
5.	$N(0, 1)$	[-3.6943, 4.26]	$[-9.97, 10.16]_a$ $[-9.97, 10.16]_b$ $[-8.82, 9.75]_c$ $[-9.97, 10.16]_d$	12.17 12.17 10.62 12.18	False_t False_t True_t -
6.	$N(0, 4)$	[-14.26, 18.00]	$[-42.90, 40.43]_a$ $[-42.90, 40.43]_b$ $[-38.41, 38.90]_c$ $[-42.90, 40.43]_d$	-51.05 -51.05 -45.05 -51.05	True_w -
7.	$\chi^2(2)$	[0.0000, 21.93]	$[-17.85, 24.33]_a$ $[-17.85, 24.33]_b$ $[-15.48, 23.44]_c$ $[0.0006, 24.33]_d$	n/a n/a n/a -2.40	n/a n/a n/a -
8.	$\chi^2(10)$	[0.7468, 33.64]	$[-37.59, 57.96]_a$ $[1.96, 57.96]_b$ $[-31.33, 26.34]_c$ $[1.08, 26.46]_d$	n/a -31.10 n/a 8.29	n/a True_t n/a -
9.	$Weibull(1)$	[0.0000, 12.23]	$[-9.56, 12.23]_a$ $[0.0021, 12.23]_b$ $[-8.19, 10.47]_c$ $[0.0012, 12.23]_d$	n/a 0.00 n/a 0.00	n/a False_w n/a -
10.	$Weibull(5)$	[0.1209, 1.60]	$[-1.20, 3.22]_a$ $[0.2351, 1.47]_b$ $[-1.00, 3.06]_c$ $[0.1209, 1.60]_d$	n/a 0.25 n/a 0.00	n/a - n/a n/a

- Estimated range with the subscripts SaS, SbS, and SdS stand for online bootstrap, double online bootstrap, traditional bootstrap, and online bootstrap with noise filtering estimators, respectively.

- Range error, indicated by the bold typeface, shows the minimum error. Its positive or negative values indicate whether the estimated range is narrower or wider than the true population range, respectively.

- Hypothesis test: H_0 : "There is no significant difference in the range error mean between method d and the compared method over the specified time period.

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Table 7Estimated results with chunk size $|c| = 50$ in real-world datasets.

No.	Distribution	Population range (<i>pop_range</i>)	Estimated range (<i>est_range</i>)	Range error (<i>rage_error</i>)	Reject H_0
1.	Laptop prices	[174, 6,059]	[191.90, 6,099] _a [191.90, 6,099] _b [196.29, 5,176] _c [191.90, 6,099] _d	17.90 17.90 945.14 17.90	False _w False _w True _w -
2.	Electronic sales	[20.75, 11,389]	[15.99, 8,460] _a [20.75, 8,448] _b [20.75, 8,448] _c [18.44, 8,456] _d	2931.34 2948.50 2948.50 2936.60	False _t True _t True _t -
3.	E-commerce sales	[100, 9996]	[103, 10,003] _a [290, 9,613] _b [139, 9,975] _c [101, 10,005] _d	-5 572 59 -9	True _t True _w True _t -
4.	World tourism economy	[0.1578, 25.86]	[0.5553, 25.86] _a [0.5567, 25.86] _b [0.6324, 25.48] _c [0.4650, 25.86] _d	2.73 2.73 3.19 2.64	True _w True _w True _w -

- Estimated range with the subscripts a, b, and d stand for online bootstrap, double online bootstrap, traditional bootstrap, and online bootstrap with noise filtering estimators, respectively.

- Range error, indicated by the bold typeface, shows the minimum error. Its positive or negative values suggest whether the estimated range is narrower or wider than the true population range, respectively.

- Hypothesis test: H_0 : "There is no significant difference in the mean range error between Method d and the compared method over the specified time period."

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$c_1 = \{5, 12, 8, 18\}$ $c_2 = \{3, 9, 11, 17\}$ $c_3 = \{1, 4, 19, 20\}$

t = 1: get c_1 so $D = \{5, 8, 12, 18\}$ L = 5, R = 18
expand L and R: L = 3, R = 19 by bootstrapping

t = 2: get c_2 so $D = \{3, 9, 11, 17\}$ L = 3, R = 19
expand L only: L = 3, R = 19
 $3 \leq 3, 9, 11, 17 \leq 19$ no bootstrapping

t = 3: get c_3 so $D = \{1, 4, 19, 20\}$ L = 3, R = 19
min(D) = 1 < L: expand L = 1
max(D) = 20 > R: expand R = 20
L is minimum. Bootstrapping R = 21

Figure 1: An example of studied problem.

Table 8

Estimated results with the chunk size $|c| = 100$

in real-world datasets.

No.	Distribution	Population range (<i>pop_range</i>)	Estimated range (<i>est_range</i>)	Range error (<i>rage_error</i>)	Reject H_0
1.	Laptop price	[174, 6,059]	[191.90, 6,099] _a [191.90, 6,099] _b [195.97, 5,099] _c [191.90, 6,099] _d	17.90 17.90 1021.86 17.90	False _t False _t False _t -
2.	Electronic sales	[20.75, 11,389]	[18.71, 8,473] _a [19.19, 8,448] _b [20.75, 8,448] _c [19.88, 8,476] _d	2921.62 2946.94 2948.50 2920.24	True _t True _t True _t -
3.	E-commerce sales	[100, 9996]	[105, 9,997] _a [290, 9,973] _b [137, 9,977] _c [102, 10,000] _d	4 212 56 -3	False _w False _w True _t -
4.	World tourism economy	[0.1578, 25.86]	[0.5500, 25.86] _a [1.5399, 25.86] _b [0.6268, 25.55] _c [0.5534, 25.86] _d	2.72 3.72 3.12 2.73	False _t True _w True _w -

- Estimated range with the subscripts a, b, and d stand for online bootstrap, double online bootstrap, traditional bootstrap, and online bootstrap with noise filtering estimators, respectively.
- Range error, indicated by the bold typeface, shows the minimum error. Its positive or negative values suggest whether the estimated range is narrower or wider than the true population range, respectively.
- Hypothesis test: H_0 : "There is no significant difference in the mean range error between Method d and the compared method over the specified time period."

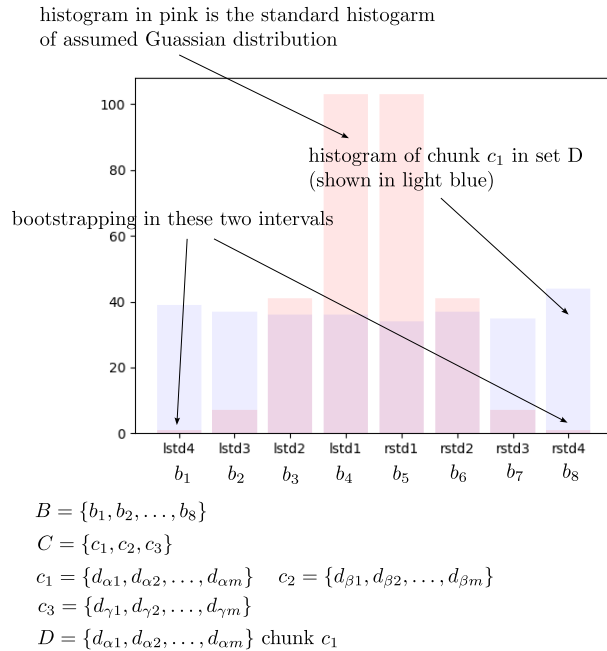


Figure 2: An example of standard histogram of probabilistic distribution (shown in pink) versus the data histogram of incoming data distribution (shown in blue color).

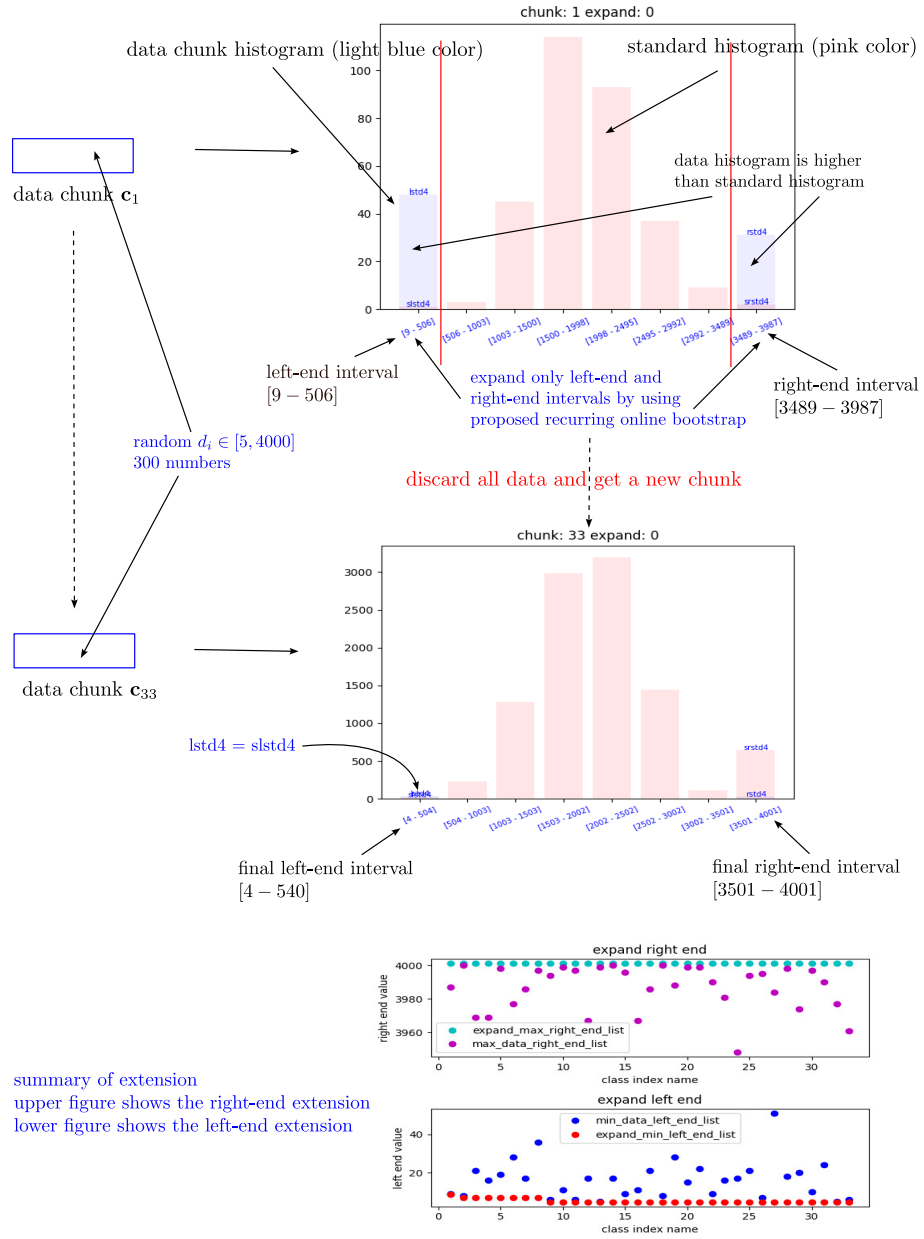
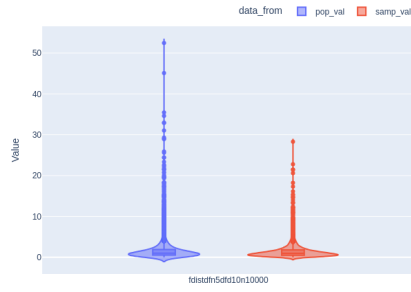
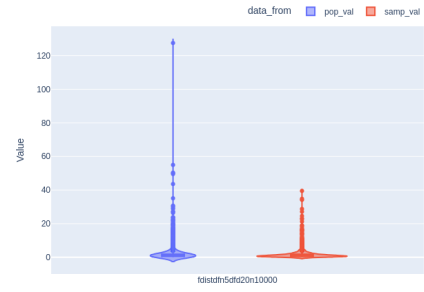


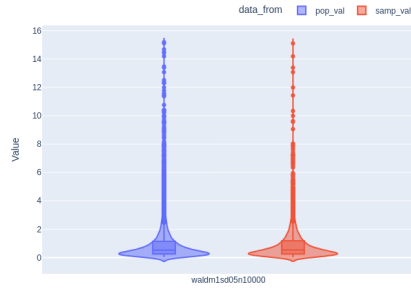
Figure 3: Proposed concept of recurring online bootstrapping method.



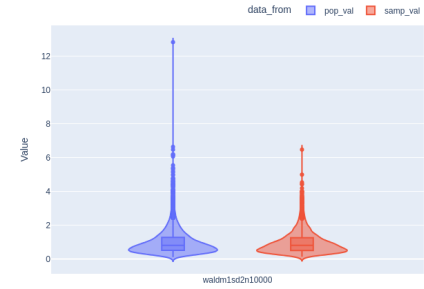
(a) $F(\nu_1 = 5, \nu_2 = 10)$



(b) $F(\nu_1 = 5, \nu_2 = 20)$



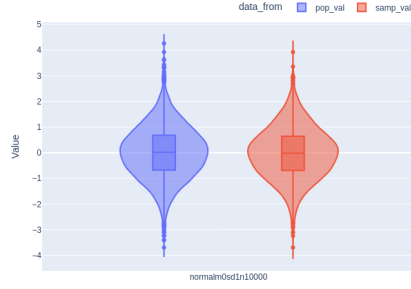
(c) $Wald(\mu = 1.0, \lambda = 0.5)$



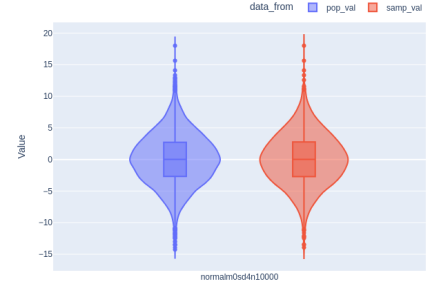
(d) $Wald(\mu = 1.0, \lambda = 2.0)$

Figure 4: Simulated population and sample distribution for F and $Wald$ distributions with $N = 10,000$

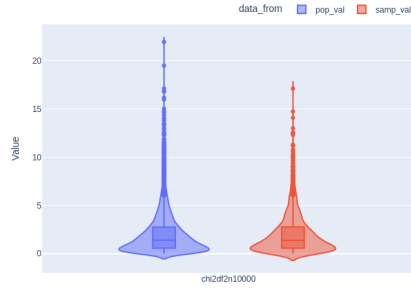
Note: The blue violin plots shows the distribution of the complete simulated population ($N = 10,000$), while the red one shows the distribution of 30% of random sample drawn from the population



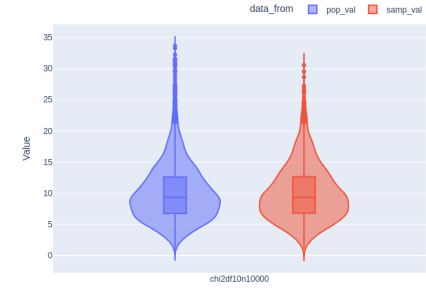
(a) $N(\mu = 0, \sigma^2 = 1)$



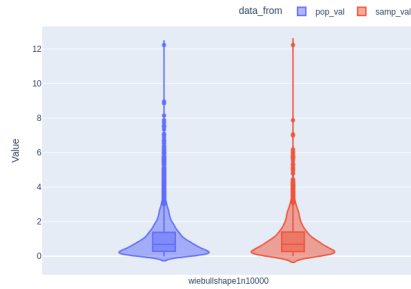
(b) $N(\mu = 0, \sigma^2 = 16)$



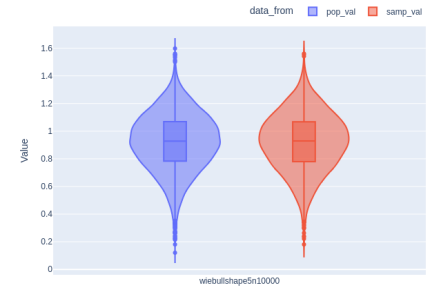
(c) $\chi^2(\nu = 2)$



(d) $\chi^2(\nu = 10)$



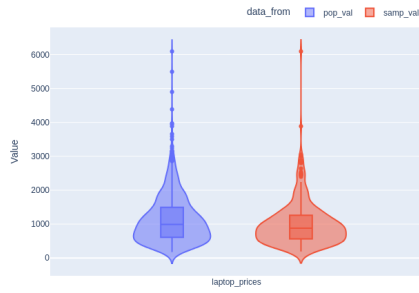
(e) $Weibull(shape = 1.0)$



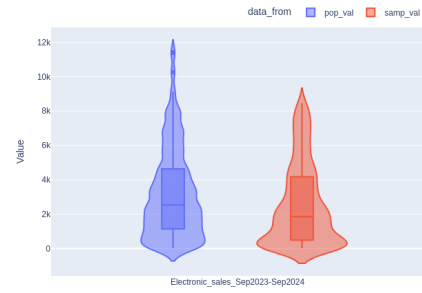
(f) $Weibull(shape = 5.0)$

Figure 5: Simulated population and sample distribution for $Normal$, χ^2 , and $Weibull$ distributions with $N = 10,000$

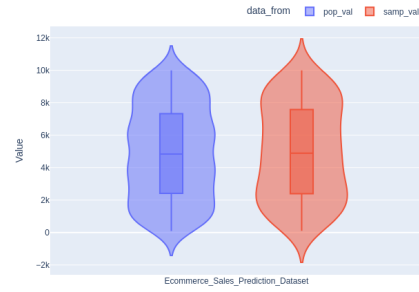
Note: The blue violin plots shows the distribution of the complete simulated population ($N = 10,000$), while the red one shows the distribution of 30% of random sample drawn from the population



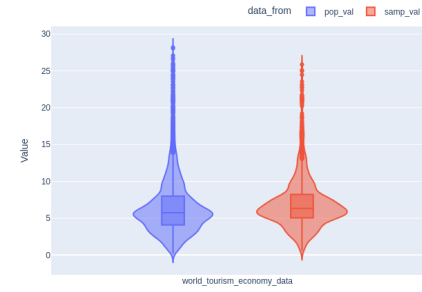
(a) Laptop prices



(b) Electronic sales



(c) E-commerce sales



(d) World tourism economy

Figure 6: Real-world population data distribution for laptop prices, electronic sales, e-commerce sales, and world tourism economy.

Note: The blue violin plots shows the distribution of the complete simulated population ($N = 10,000$), while the red one shows the distribution of 30% of random sample drawn from the population